

Introduction to Contract Theory and Auctions

Math Appendix: Facts about concave/convex functions, solving ODEs by the Integrating Factor, and Affiliated Random Variables.

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Concave functions

Some facts about concave functions:

- ① An increasing and concave function must be continuous (but not necessarily differentiable.)
- ② *Jensen's inequality.* If u is concave, then for any finite collection of numbers $(\alpha_k)_{k=1,\dots,K}$ that add up to 1, $\sum_{k=1}^K \alpha_k u(x_k) \leq u\left(\sum_{k=1}^K \alpha_k x_k\right)$. Alternatively, if F is a CDF, if u is concave

$$\int u(s) dF(s) \leq u\left(\int s dF(s)\right).$$

- ③ *Three Strings Lemma:* For any $a < b < c$, we have

$$\frac{u(c) - u(b)}{c - b} \leq \frac{u(c) - u(a)}{c - a} \leq \frac{u(b) - u(a)}{b - a}.$$

- ④ If u is twice differentiable, then for any $a < c$, $u'(a) \geq u'(c)$, and thus $u''(x) \leq 0$ for all x .

Some facts about convex functions:

- ① An increasing and convex function must be continuous (but not necessarily differentiable.)
- ② *Jensen's inequality.* If u is convex, then for any finite collection of numbers $(\alpha_k)_{k=1,\dots,K}$ that add up to 1, $\sum_{k=1}^K \alpha_k u(x_k) \geq u\left(\sum_{k=1}^K \alpha_k x_k\right)$. Alternatively, if F is a CDF, if u is convex (concave)

$$\int u(s) dF(s) \geq u\left(\int s dF(s)\right).$$

- ③ *Three Strings Lemma:* For any $a < b < c$, we have

$$\frac{u(c) - u(b)}{c - b} \geq \frac{u(c) - u(a)}{c - a} \geq \frac{u(b) - u(a)}{b - a}.$$

- ④ If u is twice differentiable, then for any $a < c$, $u'(a) \leq u'(c)$, and thus $u''(x) \geq 0$ for all x .

Solving ODEs by the Integrating factor

- Method to solve linear differential equations that have the form:

$$b'(x) + b(x)Q(x) = R(x)$$

- Notice that in the equation above, the LHS has a slight similarity to the derivative of a product $(bp)' = b'p + bp'$.
- Suppose that we multiply the LHS by a function $P(x)$. Then it becomes

$$b'(x)P(x) + b(x)P(x)Q(x).$$

- If we can choose $P(x)$ such that $P'(x) = P(x)Q(x)$, then the LHS can be transformed into the derivative of the product $b(x)P(x)$:

$$\begin{aligned}(b(x)P(x))' &= b'(x)P(x) + b(x)P'(x) \\ &= b'(x)P(x) + b(x)P(x)Q(x) \\ &= (b'(x) + b(x)Q(x))P(x) \\ &= R(x)P(x)\end{aligned}$$

Solving ODEs by the Integrating factor

- $P(x)$ is called the **integrating factor**.
- To find P such that $P' = PQ$ note that

$$(\ln P)' = \frac{P'}{P} = Q$$

- Thus $P(x) = \exp\left(\int Q(s)ds\right)$.
- For the original differential equation, we have:

$$\int (b(s)P(s))' ds = \int R(s)P(s)ds$$

- If we add a boundary condition, e.g. $b(0) = b_0$ we get

$$b(x)P(x) - b_0P(0) = \int_0^x R(s)P(s)ds \Rightarrow b(x) = \frac{\int_0^x R(s)P(s)ds + b_0P(0)}{P(x)}$$

Affiliated random variables

- In the interdependent values model, we allow the joint distribution of the signals $f(\mathbf{X})$ to be correlated, thus, it is not a product of the marginal densities.
- **Affiliation** is a strong form of positive correlation.
 - Intuitively, it means that if a subset of the signals X_i 's are all large, then this makes more likely that the remaining X_j 's are also large.

Affiliation

Suppose the random variables $X_1, X_2, \dots, X_n$ are distributed on some product of intervals $\mathcal{X} \subseteq \mathbb{R}^n$ according to the joint density function f . The random vector $\mathbf{X} = (X_1, \dots, X_n)$ is said to be **affiliated** if for all $\mathbf{x}', \mathbf{x}'' \in \mathcal{X}$,

$$f(\mathbf{x}' \vee \mathbf{x}'')f(\mathbf{x}' \wedge \mathbf{x}'') \geq f(\mathbf{x}')f(\mathbf{x}''),$$

where $\mathbf{x}' \vee \mathbf{x}'' = (\max(x'_1, x''_1), \max(x'_2, x''_2), \dots, \max(x'_n, x''_n))$ is the **component-wise maximum** of $\mathbf{x}'$ and $\mathbf{x}''$ and $\mathbf{x}' \wedge \mathbf{x}'' = (\min(x'_1, x''_1), \min(x'_2, x''_2), \dots, \min(x'_n, x''_n))$ is the **component-wise minimum** of $\mathbf{x}'$ and $\mathbf{x}''$

Affiliation: Remarks and characterizations

- A function g is said to be **supermodular** if $g(\mathbf{x}' \vee \mathbf{x}'') + g(\mathbf{x}' \wedge \mathbf{x}'') \geq g(\mathbf{x}') + g(\mathbf{x}'')$. Thus f is **affiliated** if and only if $\ln f$ is super modular, or **log-supermodular**.
- If the density $f : \mathcal{X} \rightarrow \mathbb{R}_+$ is strictly positive in the interior of $\mathcal{X}$ and twice continuously differentiable, then affiliation is equivalent to

$$\frac{\partial^2}{\partial x_i \partial x_j} \ln f \geq 0, \quad \text{for all } i \neq j.$$

Symmetry

Consider the domain $\mathcal{X} = [0, \omega]^n$. A function $f : \mathcal{X} \rightarrow \mathbb{R}$ is **symmetric** in its n variables if for every permutation^a σ of the set $\{1, \dots, n\}$, and for every $\mathbf{x} = (x_1, \dots, x_n) \in \mathcal{X}$,

$$f(x_1, x_2, \dots, x_n) = f(x_{\sigma(1)}, x_{\sigma(2)}, \dots, x_{\sigma(n)}).$$

^aA permutation $\sigma : S \rightarrow S$ is a **bijection**: one-to-one and onto. Example: $S = \{1, 2, 3, 4, 5, 6\}$, $\sigma(1) = 2, \sigma(2) = 6, \sigma(3) = 5, \sigma(4) = 4, \sigma(5) = 3, \sigma(6) = 1$. This can be denoted as

$$\sigma = \begin{pmatrix} 1 & 2 & 3 & 4 & 5 & 6 \\ 2 & 6 & 5 & 4 & 3 & 1 \end{pmatrix}.$$

- Suppose that the vector $\mathbf{X}$ has a symmetric distribution.
- Define $Y_1, Y_2, \dots, Y_{n-1}$ to be the largest, second largest, etc. from among $X_2, X_3, \dots, X_n$, i.e., the sequence of order statistics with sample size $(n - 1)$.
- If g denotes the joint density of $X_1, Y_1, \dots, Y_{n-1}$, then

$$g(x_1, y_1, y_2, \dots, y_{n-1}) = (n - 1)! f(x_1, y_1, \dots, y_{n-1}) \quad y_1 \geq y_2 \geq \dots \geq y_{n-1} \quad 0 \text{ otherwise.}$$

- Then, it follows that if $X_1, X_2, \dots, X_n$ are symmetrically distributed and affiliated, then $X_1, Y_1, Y_2, \dots, Y_{n-1}$ are also affiliated.

- Suppose that X and Y have a joint density $f : [0, \omega]^2 \rightarrow \mathbb{R}_+$. If X and Y are affiliated, then for all $x' \geq x$ and $y' \geq y$

$$f(x', y)f(x, y') \leq f(x, y)f(x', y')$$

- The above display is equivalent to

$$\frac{f(x, y')}{f(x, y)} \leq \frac{f(x', y')}{f(x', y)}$$

- Let $F(\cdot|x) = F_Y(\cdot|X = x)$ denote the conditional distribution of Y given $X = x$, and let $f(\cdot|x) = f_Y(\cdot|X = x)$ denote the corresponding pdf. Then the previous display is equivalent to

$$\frac{f(y'|x)f(x)}{f(y|x)f(x)} \leq \frac{f(y'|x')f(x')}{f(y|x')f(x')} \Rightarrow \frac{f(y|x')}{f(y|x)} \leq \frac{f(y'|x')}{f(y'|x)}$$

Likelihood ratio and Affiliation.

For the joint density $f : [0, \omega]^2 \rightarrow \mathbb{R}_+$, we define the **likelihood ratio** for all $x' \geq x$ as

$$\frac{f(\cdot|x')}{f(\cdot|x)}$$

Thus, affiliation: $\frac{f(y|x')}{f(y|x)} \leq \frac{f(y'|x')}{f(y'|x)}$ implies that the likelihood ratio is increasing in y for all $x' \geq x$.

The above is called the **monotone likelihood ratio property**.

We say that $F(\cdot|x')$ dominates $F(\cdot|x)$ in **terms of the likelihood ratio**.

Affiliation with two variables and stochastic orders.

Domination in terms of the likelihood ratio is the strongest stochastic order and it implies other stochastic orders.

That is, for all $x' \geq x$:

- $F(\cdot|x')$ dominates $F(\cdot|x)$ in terms of the **hazard rate**:

$$\lambda(y|x') = \frac{f(y|x')}{1 - F(y|x')} \leq \frac{f(y|x)}{1 - F(y|x)} = \lambda(y|x) \quad \forall y$$

- $F(\cdot|x')$ dominates $F(\cdot|x)$ in terms of the **reverse hazard rate**:

$$\sigma(y|x') = \frac{f(y|x')}{F(y|x')} \geq \frac{f(y|x)}{F(y|x)} = \sigma(y|x) \quad \forall y$$

- $F(\cdot|x')$ dominates $F(\cdot|x)$ in terms of **first-order stochastic dominance**:

$$F(y|x') \leq F(y|x) \quad \forall y$$

- If X and Y are affiliated, then for all y , $F(y|\cdot)$ is non-increasing in x implies that the conditional expectation of Y conditional on $X = x$ $E[Y|X = x]$ is a non-decreasing function of x .
- If γ is a non-decreasing function, then $E[\gamma(Y)|X = x]$ is a non-decreasing function of x .
- A more general statement: if $X_1, X_2, \dots, X_n$ are affiliated and γ is a non-decreasing function, then for all i

$$E[\gamma(\mathbf{X})|x'_1 \leq X_1 \leq x''_1, x'_2 \leq X_2 \leq x''_2, \dots, x'_n \leq X_n \leq x''_n]$$

is a non-decreasing function of x'_i and x''_i .